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Parametric study on beta-type Stirling engine

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ABSTRACT

In this work, a parametric study on a β-type Stirling engine with no regenerator was conducted numerically using ANSYS fluent 14.5 software. The three parameters that were studied are; initial charge pressure, thermal boundary condition; and three different types of working fluids (Air, He and H_2). Variable thermal properties of these gases were adopted to get more realistic results. The results include a comparison of the amount of heat transfer, power output, and thermal efficiency. It was found that the best engine performance is achieved when H_2 gas is used as working fluid. Moreover, results revealed that each of the power output and the efficiency has different optimum charge pressure. Additionally, it was found that there is a small variation in the pressure across the engine chambers, which results in miss matching between the net heat transfer rates and power output calculated from PV-diagram. This error is higher when the air is used as working fluid, especially at high charge pressure.

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1. Introduction

Stirling engine is one of the most promising innovations that is expected to help in solving energy conversion problems. It attracts researchers in industry and academic fields due to its interesting characteristics. It is classified as an external combustion engine. It can utilize various types of heat resources. For instance geothermal heat, industrial waste heat and even heat from fossil fuel combustion. However, when solar energy is utilized, the system will be environmentally friendly with no more carbon dioxide produced. In addition, it has a quiet operation with a low level of noise. The most important feature of the Stirling engine is its ability to operate in the absence of oxygen. That makes it an area of research for outer space applications. Another main application of Stirling engine is by using solar parabolic mirrors focused on the hot chamber of Stirling engine to generate electricity. So it is useful for countries with abundant solar energy.

There are five methods used to design and predict the performance of Stirling engine; zero-order analysis, first-order analysis, second-order analysis, third-order analysis (or nodal design method) and fourth order analysis (or CFD simulation) [1,2]. The term "order" used here is a kind of classification according to the complexity of the model that increases as the order increases.

Mahkamov [3] is one of the leaders in numerical modeling of Stirling engine. He simulated a prior manufacturing prototype of gamma-type Stirling engine. The engine was already designed using first-order model and the power expected was far less than the actually obtained power. His investigation showed that there are two major factors limiting the power output; firstly, the hydraulic losses in the regenerator and, secondly, the large dead volume. Consequently, he modified the prototype by eliminating some of the dead volumes through converting Stirling engine configuration from γ-type to α-type. Ibrahim [4] developed a 2-D-axisymmertic CFD model to simulate the experimental test rig developed by Jiang and Simon [5]. The original experimental (3-D) geometry had radially oriented slots in the heater. So it could not be simulated as 2-D-axisymmetric. However, using same hydraulic diameter, Ibrahim [4] found a good agreement between experimental and CFD results in the region far away from the heater. He showed that most of the pressure drop occurs in the regenerator at 90° and 270° crank angles where the piston and flow had the highest velocity. His results showed an error in the heat transfer coefficient fluctuating between 23% and 75%. Costa et al. [6] studied numerically the pressure drop in a 3-D wound woven wire matrix for Stirling engine. They studied two various regenerators; stacked wire and wound woven wire matrix. Their study focused on the effect of porosity. They derived a correlation to estimate the pressure drop friction factor for stacked woven wire. They included the wire mesh diameter in the correlation as well as *Re* number because it showed an effect on the friction factor. Another investigation of a compact porous-sheets heat exchanger was conducted by Li et al. [7]. They run a 3-D simulation for α-type Stirling

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Nomenclature

a, b, c, d	constants
c_p	specific heat at constant pressure (kJ kg^{-1} K^{-1})
g	gravitational acceleration (m s^{-2})
k	thermal conductivity (W m^{-1} K^{-1})
l_1, l_2, l_3, l_4	lengths of the linkages of the rhombic drive mechanism (m)
l_d	height of displacer (m)
n	normal direction to a plane
p	pressure (Pa or bar)
q	heat flux (W m^{-2})
r_1, r_2	radius of displacer and cylinder (m)
t	time (s)
u, v	velocity components in x- and r-direction (m s^{-1})
$\tilde{u}, \tilde{v}$	relative velocity in x- and r-direction (m s^{-1})
u_c, v_c	moving frame velocity components in x- and r-direction (m s^{-1})
u_p, u_d	piston and displacer velocities (m s^{-1})
x, r	cylindrical coordinates in x- and r-direction
G	regenerative channel gap (m)
L	distance between two gears (m)
L_{dt}	length from the linkage $l_4$to top surface of the displacer (m)
L_{pt}	height from the linkage $l_1$to top surface of the piston (m)
L_t	height of the engine (m)
Q_{in}, Q_{out}	heat absorb and rejected from the Stirling engine (W)
Q_i	total heat transfer during step i (W)
R	gas constant (J kg^{-1} K^{-1})
R_d	offset distance from the crank to the center of gear (m)
T, T_C, T_H	temperature, cold-end temperature, and hot-end temperature (K)
V	volume (m^3)
W_{out}	power output (W)

Greek symbols

δ	solid wall thickness (cylinder wall) (m)
η	thermal efficiency (%)
μ	viscosity (kg m^{-1} s^{-1})
θ	crank angle (degree or radian)
ρ	density (kg m^{-3})
ω	rotational speed (rad s^{-1})

Subscript

f	fluid

engine in which the piston motion is controlled by Ross mechanism. The engine was charged with Helium at atmospheric pressure. They concluded that in a comparison between the conventional wire mesh regenerator and the porous sheet regenerator, the later has the lowest entropy generation rate. Thus, the use of the porous sheet regenerator results in high thermal efficiency and high power output. Chen et al. [8] developed a CFD code to study the characteristics of heat transfer for 3-D γ-type Stirling engine driven with crankshaft mechanism. Their studies were conducted for very simple geometry without a regenerator for the sake of simplifications. They found differences between their results and those obtained by the second order method especially in the heat transfer rates. Moreover, they discovered a highly non-uniform temperature distribution within the engine during the cycle which reflects an inaccuracy of some of the one-dimensional models. Alfarawi et al. [9] adopted an ideal adiabatic model in order to estimate the performance of a β-type Stirling engine. Besides, they carried out a CFD investigation to optimize the geometry of the heater and the cooler. The ideal thermal efficiency was found to be 40% with a power output of 1.5–2 kWe. Another numerical and experimental study of Stirling engine regenerator was conducted by Tutar et al. [10]. They studied the pressure drop losses in wound woven wire matrix regenerator. They tested a porosity range between 52% and 72% and Reynolds number (Re) ranging from 5 to 50 in the laminar flow regime. Then they came up with a correlation related the friction coefficient factor with Re.

Furthermore, Costa et al. [11] developed a thermal non-equilibrium porous media model for Stirling engine. They found that the stacked configuration has better overall performance compared with the wound woven wire matrices. Chen et al. [12] investigated the impacts of moving regenerator on the performance of β-type Stirling engine driven by rhombic mechanism. They studied the effect of different values of porosity and discovered that despite the fact that the pressure loss is proportional with the porosity, the increase of porosity improves the overall engine performance in terms of net power output and thermal efficiency. Mahkamov and Ingham [13,14] investigated the performance of complex geometry of an alpha-type Stirling engine. Their results showed that a second-order analysis produces almost double the value of CFD analysis in terms of power. NASA also paid attention to developing and improving Stirling engine. One of the fastest simulations of the whole Stirling engine was performed by Dyson et al. [15]. They simulated a free-piston configuration of Stirling engine using axisymmetric. A comparison of their results with an actual performance data showed that the axisymmetric simulation is more accurate than Sage code results. Sage code is one-dimensional analysis tool supports simulation and optimization of spring-mass-damper resonant system and Stirling engine cycle [16]. It is worth mentioning here that they set the boundary condition for the cold-end temperature in their simulation lower than temperature value in actual data by about 14 °C. Because there might be some losses in the experiment, it needs to be overcome by increasing the source temperature a little bit. The outcomes of heat supply, heat rejected and generated power of the simulation have good agreement with the experimental data. Lin et al. [17] conducted a simulation analysis of 3-D free-piston Stirling engine, which used a helium as the working fluid. Unfortunately, their results in terms of heat added to the cycle did not match the experimental data. The CFD model reflected a high amount of heat input compared to the experimental data. Costa et al. [18] studied heat transfer in a 3-D wound woven wire matrix of a Stirling engine regenerator. Firstly, they developed a 3-D model for stack woven and validated the obtained correlation of heat transfer coefficient against empirical relationship of Tanaka et al. [19] and Gedeon and Wood's work [20]. After that, they extended the model for wound woven wire matrices. They considered the flow to be laminar in the low Re number range and turbulent at higher Re number range. Their results reflected a value of Nu number lower by 20% for wound woven matrices compared with that of stack woven matrices for $20 < Re < 400$. Moreover, they did not notice a significant impact of volumetric porosity on Nu due to the small range of volumetric porosity (0.6–0.68). Therefore, they suggested studying it in further investigation. Salazar and Chen [21] investigated the characteristics of rhombic-drive beta-configuration Stirling engine working with air as the working fluid. The problem was treated as axisymmetric laminar and thermal properties along with the viscosity were assumed to be constant. In addition, it was proposed

that there was no regenerator in the model. Thus, a small value of thermal efficiency (about 14%) was obtained while the corresponding Carnot efficiency reaches nearly 62.5%. Their results emphasized the poor assumption of constant heat transfer coefficient in the second-order analysis because of non-uniform temperature within Stirling engine.

The purpose of this work is to perform a parametric study on a β-type Stirling engine in order to find the impact of the major factors on its performance. There are three factors are studied in the present work, which are namely; working fluid type, initial charge pressure and thermal boundary conditions. This comprehensive study was carried out because it was noticed that there is a shortage in such parametric study in the literature. Besides, this research will provide more information that would help more in understanding the characteristics of Stirling engine.

2. Problem formulation

2.1. Geometry description

In this CFD simulation, a β-type Stirling engine driven by a rhombic mechanism is considered. It is the same geometry as that used by Salazar and Chen work [21]. It consists of three zones; compression zone, expansion zone, and narrow zone, which connects the expansion and compression zones as shown in Fig. 1.

In general, this narrow channel is filled with regenerator materials. However, the presently considered Stirling engine configuration does not include a regenerator. It can be seen also from Fig. 1 that it is easy to notice that the problem could be solved as axis-symmetric geometry. So that, only half of the domain is simulated. The dimensions of the simulated geometry are reported, hereunder, in Table 1.

This parametric study includes three types of working fluid; Air, Helium, and Hydrogen. Each one of these fluids was tested for a range of charge pressure between 1 bar and 5 bar to obtain the optimum value for the generated power and the efficiency except for hydrogen, its test was extended up to 7 bar to get the optimum point. The effect of initial charge pressure is conducted at minimum wall temperature (T_C) of 300 K and maximum wall temperature (T_H) of 800 K. The external wall is exposed to a variable temperature, which is function of axial x-coordinates:

$$T(x) = \begin{cases} T_C\ (\mathrm{K}), & \text{if } x \leqslant 0.08\ \mathrm{m} \\ T_C + \frac{x-0.08}{0.158-0.08}(T_H - T_C)\ (\mathrm{K}), & \text{if } x > 0.08\ \mathrm{m} \end{cases} \quad (1)$$

Additionally, the effect of T_H in Eq. (1) is examined for a range of 800–1300 K while T_C and the initial charge pressure were kept at 300 K and 1 bar, respectively. Also, the range of maximum and minimum temperature was shifted to a higher level by a step of 100 K until T_C reach up to 800 K and T_H reach value of 1300 K. This is to see which operation range is the best for this type of Stirling engines. For all the cases under study, the Stirling engine is set to rotate at a speed of 1800 rpm. Also, the thermal properties of the three working fluids were taken as variable with the temperature.

The cylinder wall has a thickness of 1 mm and it is made of steel. It has properties of: density (7840 kg/m^3), thermal conductivity (43 m^{-1} K^{-1}) and specific heat (450 J kg^{-1} K^{-1}).

2.2. Governing equations

The transport equations describing the flow fields and heat transfer phenomena inside Stirling engine are embodied mathematically by transient axisymmetric compressible Never-Stokes equations, conservation of energy equation, conservation of mass and ideal gas equations. Those equations can be written in cylindrical coordinates as:

The continuity:

$$\frac{\partial \rho_f}{\partial t} + \frac{\partial}{\partial x}(\rho_f \tilde{u}) + \frac{1}{r}\frac{\partial}{\partial r}(\rho_f r \tilde{v}) = 0 \quad (2)$$

The momentum:

$$\begin{aligned} &\frac{\partial}{\partial t}(\rho_f u) + \frac{\partial}{\partial x}(\rho_f \tilde{u} u) + \frac{1}{r}\frac{\partial}{\partial r}(\rho_f r \tilde{v} u) = \rho_f g - \frac{\partial p}{\partial x} + \mu \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) \\ &+ \frac{u}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{\mu}{3}\frac{\partial}{\partial x}(\nabla \cdot V) + \frac{\partial}{\partial t}(\rho_f v) + \frac{\partial}{\partial x}(\rho_f \tilde{u} v) \\ &+ \frac{1}{r}\frac{\partial}{\partial r}(\rho_f r \tilde{v} v) = -\frac{\partial p}{\partial r} + \mu \frac{\partial}{\partial x}\left(\frac{\partial v}{\partial x}\right) + \frac{u}{r}\frac{\partial}{\partial r}(r\frac{\partial v}{\partial r}) \\ &+ \frac{\mu}{3}\frac{\partial}{\partial r}(\nabla \cdot V) - \frac{\mu v}{r^2} \end{aligned} \quad (3)$$

The energy:

$$\begin{aligned} &\frac{\partial}{\partial t}(\rho_f T_f) + \frac{\partial}{\partial x}(\rho_f \tilde{u} T_f) + \frac{1}{r}\frac{\partial}{\partial r}(\rho_f r \tilde{v} T_f) \\ &= -\frac{1}{c_{pf}}\left[\frac{\partial p}{\partial t} + \nabla \cdot (pV) - p(\nabla \cdot V)\right] \\ &+ \frac{k_f}{c_{pf}}\left[\frac{\partial}{\partial x}\left(\frac{\partial T_f}{\partial x}\right) + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T_f}{\partial r}\right)\right] \\ &+ \frac{\mu}{c_{pf}}\left\{2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial r}\right)^2 + \left(\frac{v}{r}\right)^2\right] + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial r}\right)^2 \right. \\ &\left. - \frac{2}{3}(\nabla \cdot V)^2\right\} \end{aligned} \quad (4)$$

Ideal gas equation:

$$pV = \rho_f R T_f \quad (5)$$

where $\tilde{u} = u - u_c$, $\tilde{v} = v - v_c$ are the relative velocity components between the fluid and moving frames with velocities of u_c and v_c in the axial direction (x) and the radial direction (r), respectively.

Initially, the gas is at 300 K and cylinder wall temperature is given by Eq. (1) which is not a function of time. The cylinder head and the piston were kept at T_H and T_C all the time, respectively. Additionally, displacer is kept insulated so no heat transfer will go through it.

Moreover, the movement of the piston and displacer are governed by a rhombic motion as shown in Fig. 1. The velocity of this kind of motion is represented by:

$$\begin{aligned} u_p(t) &= R_d \omega \left\{\cos\theta - \sin\theta\left[l_2^2 - \left(\tfrac{l}{2} - \tfrac{l_1}{2} - R_d\cos\theta\right)^2\right]^{-0.5}\left[\tfrac{l}{2} - \tfrac{l_1}{2} - R_d\cos\theta\right]\right\} \\ u_d(t) &= R_d \omega \left\{\cos\theta + \sin\theta\left[l_2^2 - \left(\tfrac{l}{2} - \tfrac{l_4}{2} - R_d\cos\theta\right)^2\right]^{-0.5}\left[\tfrac{l}{2} - \tfrac{l_4}{2} - R_d\cos\theta\right]\right\} \end{aligned} \quad (6)$$

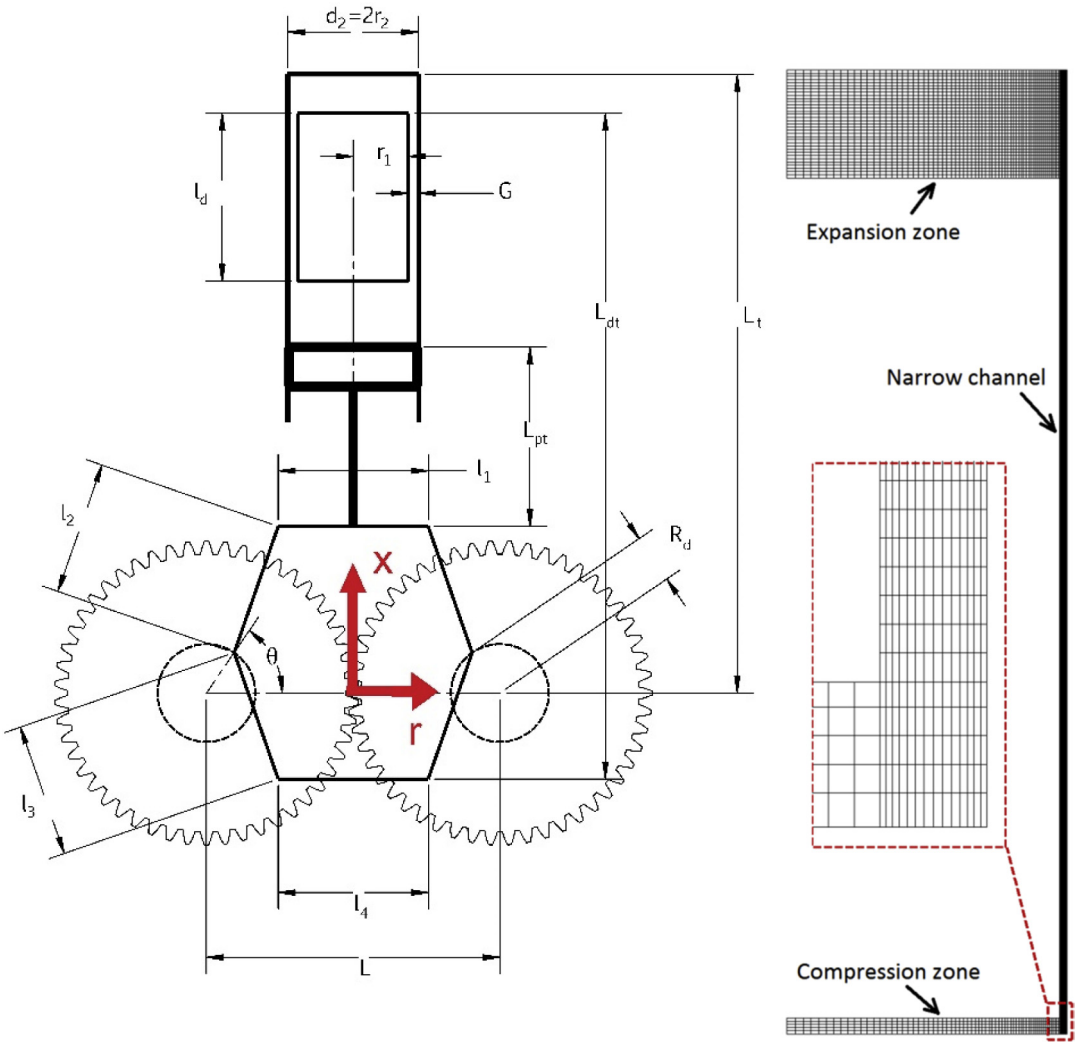
The left gear in Fig. 1 rotates in a clockwise direction, as a result, a negative value of the rotation speed (ω) was used in the above equations. The motion starts from a crank angle of zero degree.

On the other hand, the cylinder walls of Stirling engine are exposed to a conduction heat transfer in the radial direction and its energy equation for this case is given by Fourier's law:

$$q = -k\frac{\partial T}{\partial n} \quad (7)$$

Since the partial Eqs. (2)–(4) have been solved numerically, the supplied and rejected heat transfer were calculated as following:

$$\begin{aligned} Q_{in} &= \tfrac{\omega}{2\pi}\sum_{i=1}^{N} Q_i \cdot \Delta t \quad \text{if } q > 0 \\ Q_{out} &= \tfrac{\omega}{2\pi}\sum_{i=1}^{N} Q_i \cdot \Delta t \quad \text{if } q < 0 \end{aligned} \quad (8)$$



(a) Engine configuration (b) Computational domain

Fig. 1. Stirling engine β-type geometry configuration.

Table 1
Dimensions of the β-type Stirling engine in (mm).

r_1	20
r_2	20.5
G	5
l_1= l_2= l_3= l_4	18
L_t	158
L	42
L_{pt}	50.93
L_{dt}	163.74
l_d	79.46
R_d	3.5
δ	1

where Δt is time step. Q_{in}, Q_{out}, and Q_i are the total heat added, total heat rejected and total heat transfer, respectively, during the step (i). Q_i is calculated across the outer surface of the domain. The software gives the value of Q_i in terms of (W) or (J/s). It is multiplied by Δt to obtain the amount of energy transferred during the step (i). Therefore, the summation value for all energies transferred during all steps (N) of one cycle gives the total energy produced during that cycle in a unit of (J/cycle). Accordingly, it is multiplied by $(\omega/2\pi)$, which represents the number of revolutions in each second to give the heat transferred per second. While the power output calculated numerically from:

$$W_{out} = \frac{\omega}{2\pi}\sum_{i=1}^{N} 0.5(p_{i-1} + p_i)\cdot(V_i - V_{i-1}) \tag{9}$$

where p is the average pressure at specified zone, and V is the volume of that zone. The terms in the summation (in Eq. (9)) represent the average area under the P-V diagram. It is the average pressure of time step (i) and (i-1) multiply by the change of volume occurring in the same steps (i and i-1). The summation of these values for a total number of steps for one cycle (N) gives the work output per cycle (J/cycle). Then, the power output is obtained by multiplying the resultant of the summation Eq. (9) by $(\omega/2\pi)$.

3. Solution methodology

In order to solve the governing Eqs. (2)–(4), the general purpose code of ANSYS fluent 14.5 was used. The program uses the finite volume method to discretize the governing differential equations. The second order upwind finite difference scheme was adopted for continuity, momentum and energy equations. To couple the

pressure and velocity a coupled scheme was utilized because it showed a relatively fast convergence compared with SIMPLE scheme. The SIMPLE scheme uses a pressure-based segregated algorithm in which the momentum equation and the pressure correction equation are solved separately. This semi-implicit solution method results in slow convergence [22].

Furthermore, variable thermal properties were implemented in the software. An ideal gas, which is represented by Eq. (5) was used to calculate the density while the kinetic-theory was used to identify the heat capacity and the viscosity. Moreover, the effect of viscous heating dissipation (the last term in Eq. (4)) was tested and it turns out that there is a little bit difference in the results, thus, it can be neglected. Also, the effect of gravity was tested as well and it turns out to be of no impact on the engine performance. Therefore, no viscous heating nor gravity effects were included in the present parametric study.

In addition, laminar flow is assumed because the maximum *Re* based on the hydraulic diameter of the narrow channel does not exceed the laminar range for all cases under study. However, in some cases, it becomes in the transitional range (2000–4000) but it does not convert to fully turbulent conditions. The temperature profile is described by Eq. (1) and movement of the piston and the displacer (Eq. (6)) are implemented in the software through a user-defined function (UDF). The conduction heat transfer through the external walls, which is described by Eq. (7) has not been included in the computational domain. Nevertheless, the wall properties and its thickness were defined in the wall boundary condition such that the software will solve the conduction heat transfer without the need to be drawn in computational geometry.

4. Validation and mesh and time independence study

4.1. Mesh and time independence study

Before starting the simulations of Stirling engine, mesh and time dependence study were carried out. Four different mesh sizes were tested. They have a number of cells of 2740, 6869, 9600 and 15,020 cells. All meshes studied included dense mesh in the narrow space consisting of 10, 14, 16 and 20 intervals, respectively. The aspect ratio for all meshes under study was kept under 11.8. Moreover, each one of these mesh sizes was tested with three step-number per cycle of 360, 480 and 720 step/cycle which are corresponding to a time-step intervals of 9.25926×10^{-05}, 6.94444×10^{-05} and 4.62963×10^{-05} s, respectively. It was found that the mesh size of 6869 cells with 480 step/cycle (i.e. 0.94444×10^{-05} s) achieves the mesh and time-step independent solution with a deviation of 0.6% in the generated power compared with the finest mesh size with the smallest time step. In addition, it is worth mentioning here that for the simulation of the Stirling engine, the domain is scaled up by 10^{12} then it returns back to the original size in Fluent software by scaling it down by 10^{-12} in order to compensate 32-bit precision available in grid generation especially when there is fraction length as recommended by [15].

4.2. Validation against a numerical study

It is worth mentioning here that it was difficult to find previously published numerical simulation of stirling engine of the same type and configuration as the one considered in the present work for the purpose of validation. Most of the numerical work published was for α-type Stirling engine [7], γ-type Stirling engine, or β-type Stirling engine with adiabatic model [9] or with a moving regenerator [12] while the stirling engine considered in the present work is β-type Stirling engine with no regenerator. However, the work reported by Salazar and Chen [21] was for a β-type Stirling engine with boundary conditions similar to the boundary conditions considered in the present work. Therefore, after conducting the mesh independent test, the presently developed CFD model was compared with the work of Salazar and Chen [21] under the same boundary conditions for the sake of validation with previously published numerical work. The main difference between Salazar and Chen [21] work and the present work is that they utilized constant thermal properties for the working fluid. However, in the present work, variable thermal properties were used to make the problem much close to the real case. Fig. 2 shows a comparison of P-V diagram for the expansion and compression chambers. It is clear that there are good matching between the two the results of the simulation and that of Salazar and Chen [21]. However, slightly less magnitude of the power output of (7.7 W) was obtained in the present work and (11.7 W) due to the use of variable thermal properties which is the more realistic than assuming constant thermal properties.

4.3. Validation against an experimental study

In spite of having many experimental work reported the performance of stirling engine, it was difficult to find experimental work in the literature that can be taken as a reference for validating the presently developed CFD model. Most of the previously published experimental work was on α-type Stirling engine, γ-type Stirling engine [23,24], or β-type Stirling engine with different geometry or different boundary conditions [25]. Some of the experimental work reported on β-type Stirling engine was for different configuration with regenerator [25]. Moreover, many of the published experimental work on β-type Stirling engine did not report complete information about the geometrical and operating parameters that are needed as input for the CFD simulations. However, the experimental work reported on β-type Stirling engine by a group of researchers [26–28] was on a geometry and configuration similar to the geometry and configuration of the β-type Stirling engine considered in the present work. Among this group, the work reported by Aksoy and Cinar [28] represented an excellent source for information needed for validation of the present CFD simulation with experimental work. In their work [28], the performance of a beta-type Stirling engine driven by rhombic mechanism was examined. They tested two displacer shapes. The first displacer has slots grooved and the second one is without slots grooved. In

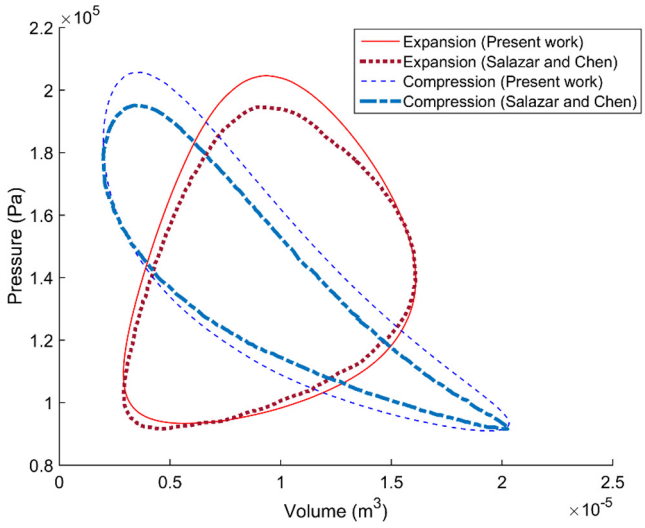


Fig. 2. P-V diagrams of expansion and compression zones comparing between the present work and Salazar and Chen's work [21].

this work, the second case (no slots grooved displacer) is selected for CFD simulation, because it can be simulated as axis-symmetry problem. Thus, it would be less computational expense compared to 3D case. The results of the CFD simulation show a power output of 26.63 W with thermal efficiency of 10.44% when laminar flow model is used. However, the experimental data shows a power output of 43.2 W. The reason for that is the flow might convert to turbulent flow since the *Re* number (based on the hydraulic diameter of the narrow channel) exceeds laminar limits in some portion of the cycle. Also, the surface roughness may play a role in converting the flow into turbulent flow. Consequently, a turbulent flow of k-ε model is examined. It shows a power output of 39.7 W with thermal efficiency of 12.7%. This shows an excellent agreement with the experimental data reported by Aksoy and Cinar [28] as (43.2 W) with a deviation of 8.1%.

5. Results and discussion

In this paper, a comprehensive operational parametric study was carried out. The investigated parameters include; initially charge pressure, operational temperature limits and the type of the working fluid. In order to illuminate the influence of these parameters on Stirling engine performance, 60 cases were examined in this study.

5.1. Effect of initial charge pressure

Fig. 3 shows the impact of initial charge pressure on heat transfer, work output, and thermal efficiency. In cases where the air is used as working fluid; the Stirling engine produces only 7.8 W at atmospherically charge pressure with an efficiency of 7.8%. As the initial charge pressure increases, the engine becomes more tending to absorb higher amount of heat. Accordingly, the power generated from the Stirling engine increased slightly to the maximum of 8.15 W at charge pressure of 1.75 bar.

At high charge pressure, the flow starts to convert into turbulent flow. Up to charge pressure of 2 bar, the *Re* number (based on the hydraulic diameter of the narrow channel) remains within a laminar region which is our concern in this work. After that, the flow goes to turbulent in a small portion of the cycle. In particular, high *Re* number occurs at the narrow channel when the displacer moves downward and the piston moves upward (i.e. when they move toward each other). Moreover, the efficiency curve starts from 7.8% at atmospheric pressure then it falls gradually until it reaches the smallest value of 5.7% at 2 bar.

On the other hand, when the working fluid was changed from air to helium gas, both of efficiency and power output were improved dramatically as shown in Fig. 4. With helium gas at atmospheric charge pressure, Stirling engine produces 14 W with an efficiency around 14.9%. These values are almost double that obtained from the air as working fluid with the same charge pressure.

Helium gas has a kinematic viscosity much greater than that of air. Thus, small *Re* number could be obtained even at large charge pressure. This allows extending the parametric study up to a charge pressure of 4 bar within the limits of laminar flow. After 4 bar charge pressure, the flow starts to convert into the turbulent regime which is not considered in this study. Among these charge pressure limits, the maximum power extracted from Stirling engine is 19.6 W. It occurs at 2.5 bar which is greater than the charge pressure that achieves optimum power output in the case of air gas.

In addition, the efficiency curve obtained from He gas has a similar pattern as that of the air but on a larger scale. It has supreme value at low charge pressure (1 bar) then it falls down gradually as charge pressure increases. It starts from 14.9% at atmospheric pressure then it drops gradually to 9.1% at 4 bar.

However, the extreme power output achieved at 2.5 bar, but it is preferred to operate the Stirling engine at 2 bar. The reason for that can easily be noticed from the efficiency and power output curves in Fig. 4. The power output will not be affected very much by reducing the charge pressure to 2 bar. On the other hand, the efficiency will increase from 12.1% to 13.3%.

Further improvement in power output was obtained by utilizing Hydrogen gas (H_2) as working fluid. The hydrogen gas is known to have the highest thermal conductivity of all gasses. As a result, a tremendous amount of heat could be absorbed as shown in Fig. 5. As the mass inside the Stirling engine increases (by increasing charge pressure) the engine becomes more able to absorb a large amount of heat due to high thermal conductivity of hydrogen. This gives Stirling engine a chance to generate large power output by using H_2 gas instead of using air or He gasses. Even at atmospheric pressure H_2 gas generates power much greater than maximum power produced by air at optimum charge pressure (17.3 W for H_2 at 1 bar and 8.15 W for air at 1.75 bar). The greatest power generated by H_2 gas is 65.2 W (with an efficiency of 7.4%) which is obtained at 12.5 bar. While at the highest thermal efficiency (11.2%) the output power is relatively low (39.3 W) which is recorded at 2.5 bar. Moreover, it can be easily noticed, from Fig. 5, that charge pressure from 7.5 bar to 12.5 bar has almost

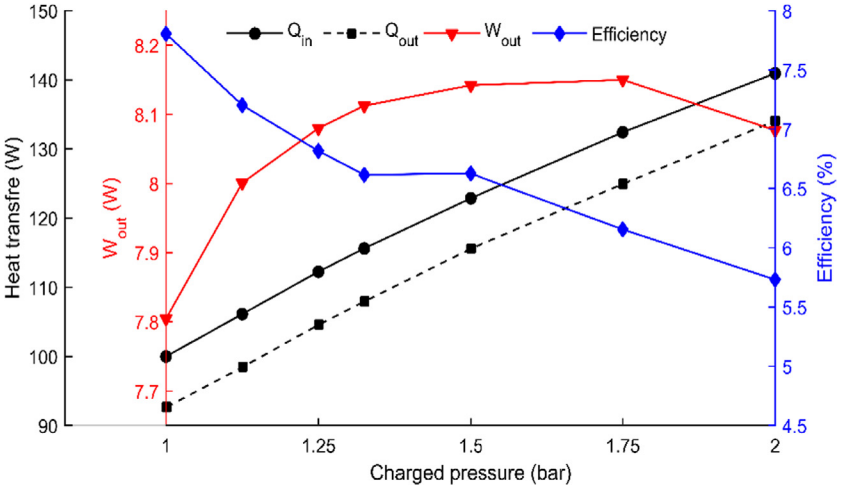


Fig. 3. Engine performance at different charge pressure and temperature limits of 300–800 K for air gas.

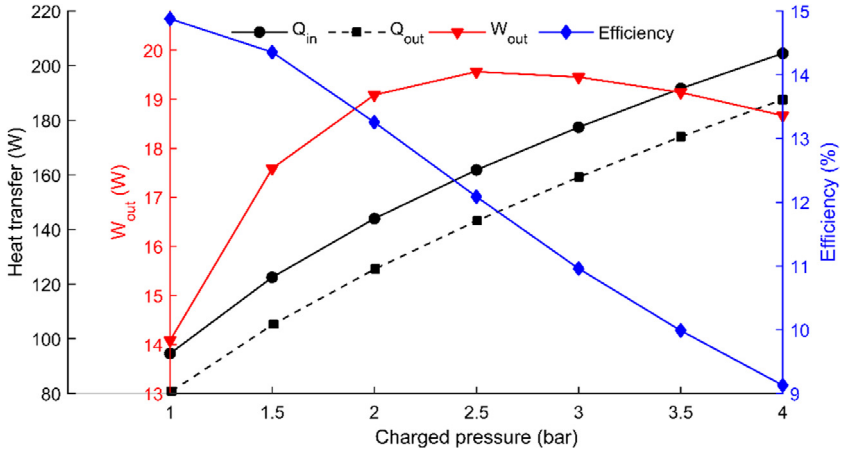


Fig. 4. Engine performance at different charge pressure and temperature limits of 300–800 K for He gas.

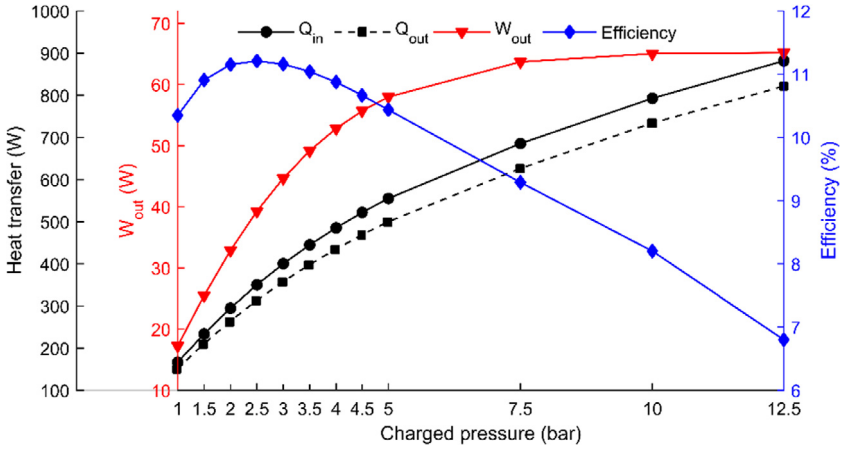


Fig. 5. Engine performance at different charge pressure and temperature limits of 300–800 K for H_2 gas.

same values of power output. However, the corresponding efficiency diminishes as charge pressure increases.

So, similar to the operation with helium gas, it would be a wise decision to operate the Stirling engine charged with H_2 gas at a pressure below the optimum one of the power output (12.5 bar) and above the optimum pressure of thermal efficiency (2.5 bar). For instance, if the engine runs at 7.5 bar instead of 12.5 bar the efficiency would improve from 7.4% to 9.3%. Thus, only 685.5 W would be required as Q_{in} instead of 881.6 W. Meanwhile, there would be a reduction in output power by only 1.5 W.

Additionally, a curve fitting was conducted for efficiency and power output in Figs. 3–5. Since the efficiency profile of hydrogen gas has a bell shape, a Gaussian function was proposed to fit the efficiency curve as following:

$$\eta_{th}(x) = a_1 e^{\frac{-(x-b_1)^2}{2c_1}} + a_2 e^{\frac{-(x-b_2)^2}{2c_2}} \quad (10)$$

where a, b and c are constants shown in Table 2 and x is the charge pressure. While the power output was represented by polynomial functions as:

$$W_{out}(x) = d_1x^4 + d_2x^3 + d_3x^2 + d_4x^1 + d_5 \quad (11)$$

where d is constant shown in Table 3 for the three working fluid.

Moreover, it has been found that at any moment of time the temperature distribution is highly non-uniform within the chambers as displayed in Fig. 6. In order to characterize the heat transfer behavior, the temperature contours were observed at various time during one complete cycle. Fig. 7 depicts the temperature contours at different time, in a cycle, for hydrogen gas at 4 bar charge pressure. Focusing on the expansion chamber, it can be noticed that as displacer comes close to the top dead center (TDC) its velocity approaches zero and the fluid velocity becomes very small. Thus, small amount of convective heat transfer occurs. Fig. 8 represents

Table 2
Coefficients of efficiency fit curve equation.

	Air	He	H_2
a_1	1.272	6.559	3.539
b_1	−1.331	−1.839	−0.8899
c_1	0.5267	1.773	1.054
a_2	6.69	9.836	9.044
b_2	−0.6116	0.05627	0.5221
c_2	5.651	4.184	3.107

Table 3
Coefficient of generated power fit curve equation.

	Air	He	H_2
d_1	−2.484	−0.2419	−0.7
d_2	15.65	0.7482	5.011
d_3	−37.06	−1.176	−13.54
d_4	39.17	0.206	16.77
d_5	−7.463	19.55	56.88

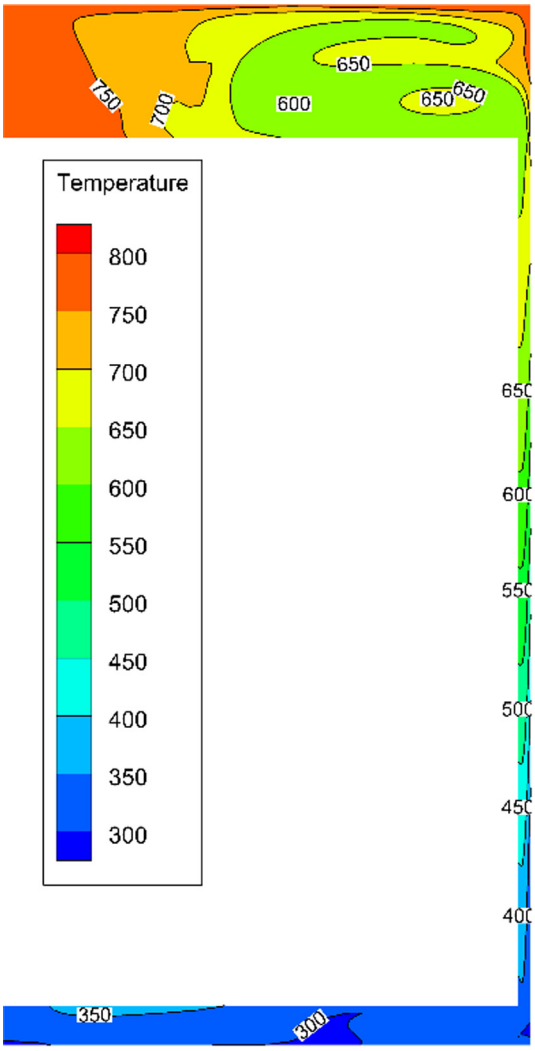


Fig. 6. Temperature contours inside Stirling engine using H_2 gas at 4 bar charge pressure and temperature limits of 300–800 K.

heat transfer on top and bottom of the cylinder. This period of no fluid motion demonstrates the declination in heat transfer on the cylinder head (i.e. the declination after step number 270 of the dashed line). Therefore, only fluid layers near the heating walls would have high temperature compared to other regions in the expansion chamber due to diffusion heat transfer mainly. Then heat transfer rate rises up quickly due to movement of the displacer toward bottom dead center (BDC) pushing the gas from compression chamber to the expansion zone. The cold jet gas coming through narrow channel would hit the cylinder head resulting in impingement heat transfer, which is known as the most effective convective heat transfer mechanism (see Fig. 7). As the displacer moves away from TDC the vortex that made by impingement of fluid on cylinder head also travels with it. Hence, the effect of impingement heat transfer degrades slowly as the vortex moved away from the cylinder head. On the other hand, monitoring heat transfer on the bottom of the cylinders shows a zig-zag fluctuation. It mainly happens due to the piston velocity variation. So that, due to non-uniform temperature distribution across the engine, the assumption of constant convective heat transfer coefficient in some second-order models could not reflect the real state of Stirling engine. Similar results were noticed in other CFD simulations of Stirling engine [8,21].

Besides non-uniform temperature distribution, there is small pressure variation within the Stirling engine at any given time during the cycle. Nevertheless, this variation is a negligible fraction of total pressure, but it leads to an error in power output calculated from Eq. (9) and power output obtained from the differences between heat added and rejected from the system (Eq. (8)). Even in Salazar and Chen [21], it seems that there was an error of 8.6% between the net heat transfer and the calculated power from P-V diagram. They got values of 82.5 W, 69.7 W and 11.7 W for heat input, heat output and generated power, respectively. Thus, the net heat transfer is 12.8 W, which is greater by 1.1 W from the calculated power. The source of this error is coming from the pressure variation across the whole engine volume. Therefore, taking the average pressure of the entire volume would lead definitely to mistake in power output value. In order to reduce this error, the complete volume should be divided into many zones. Then, the output power for each zone could be calculated separately through the p-v

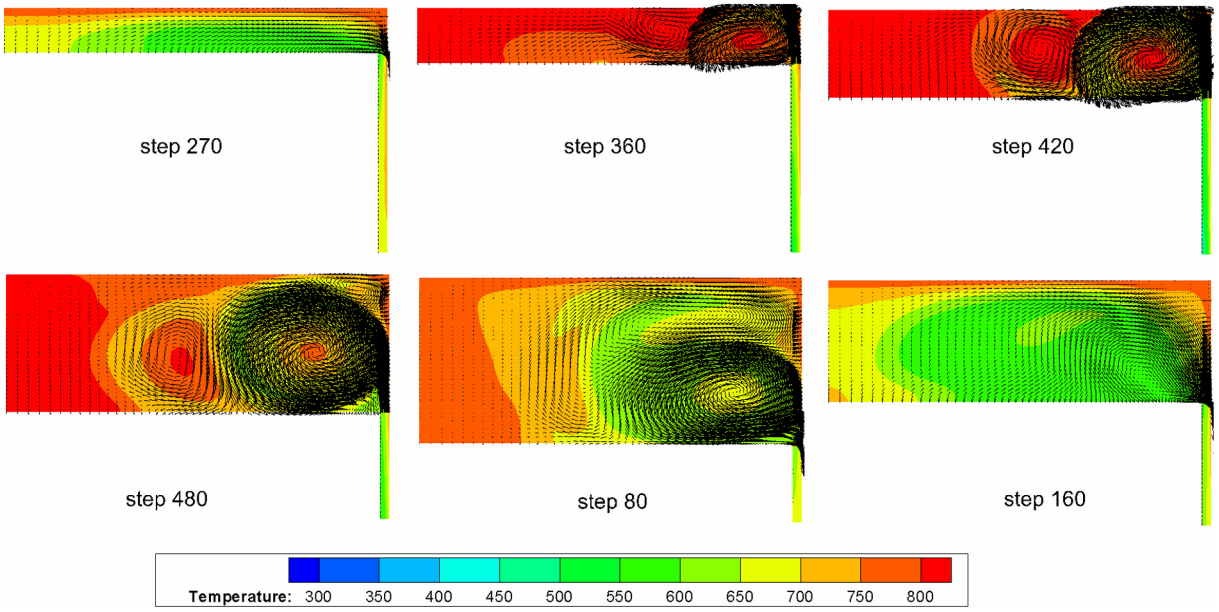


Fig. 7. Temperature contours and velocity vectors at expansion chamber at various steps for H_2 gas at 4 bar and temperature limits of 300–800 K.

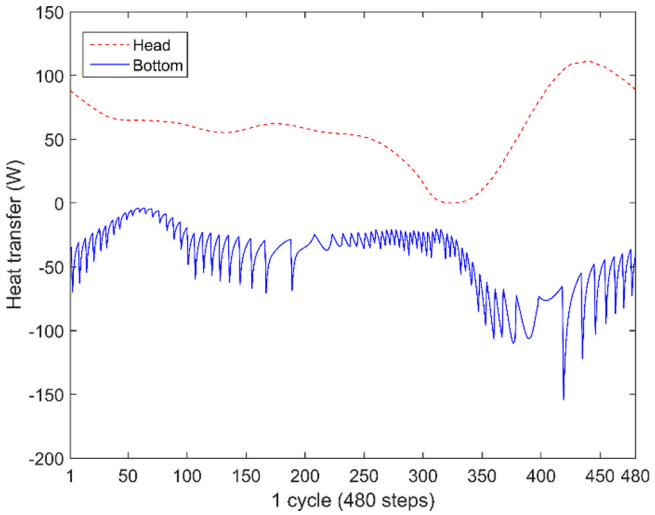


Fig. 8. Heat transfer at top and bottom of the cylinder for the case of H_2 gas at 1 bar charge pressure and temperature limits of 300–800 K.

diagram. Then the total power output is the summation of the power from these zones. In this work, the total volume was divided into three zones (expansion zone, regeneration zone, and compression zone) and the average pressure was monitored for each zone separately. Following this way, the error between net heat transfer and output power would diminish because the pressure variation within these small zones is less than pressure variation within the total volume. However, as the initial charge pressure increases the error value grows because that would make more pressure difference even in the divided small zones. In contrast, as the difference between the operational temperatures (T_H-T_C) becomes large, the pressure distribution becomes more uniform. Thus, the error between power calculated through P-V diagram (Eq. (9)) and the net heat transfer (Q_{in}-Q_{out}) (Eq. (8)) becomes small. Fig. 9 shows the error between net heat transfer and total power output calculated from the P-V diagram of an expansion and compression zones. It is clear from the figure that high error value is obtained when the average pressure is taken for the total engine volume. In contrast, the error percentage between the net heat transfer and the power output calculated from Eq. (9) is reduced as total engine volume divided into three zones (expansion zone, compression zone and narrow channel as shown in Fig. 1b) and the average pressure is taken for each zone separately.

5.2. Effect of heater temperature (T_H)

In this part of the parametric study, the gas charge pressure was taken to be atmospheric. Also, the temperature of the cold side T_C remains at 300 K while heater temperature T_H increased gradually from 700 K up to 1300 K by an increment of 100 K. Fig. 10a depicts the indicated power versus T_H for various working fluid (Air, He and H_2). It can be seen that the power increases almost linearly with increases in T_H. Additionally, in all T_H ranges under study; air registered the lowest indicated power, while hydrogen recorded the largest amount of power output. Also, Fig. 10a shows that the slope of the power output for the He gas is higher than the slope of air and H_2 gases. This indicates that at low T_H the He gas produces less power output than H_2 gas. However, as the T_H rises up, the generated power from He gas increases faster than the power generated from H_2 gas. Thus, at high T_H (for instance, at 1300 K) He and H_2 gases produce almost same amount of power output (31.9 W and 32.9 W for He and H_2 gases, respectively), while at that temperature air produces only 22 W. Furthermore, it is expected, when the T_H reaches 1500 K, using He gas would generate more power than that produced using H_2 gas. The corresponding efficiency for the three fluids is represented in Fig. 10b. Again, air shows the lowest performance while He reflects the best efficiency curve.

Comparing the efficiency curves of air and He, one can observe that increasing the difference between heater and cooler temperature has a positive impact on efficiency for both gasses. So that at heater temperature equivalent to 700 K ($\Delta T = 400$ K) both gasses gives the smallest values of efficiency (6.7% and 13.6% for air and He, respectively). Then it increases gradually as T_H increases, but with a small rate at higher temperature. Moreover, the maximum efficiency values that were obtained are 9.6% and 16.8% at 1300 K for air and He, respectively.

On the other hand, hydrogen gas did not show any significant impact of heater temperature (T_H) on the improvement of efficiency, especially at high temperature. The efficiency starts from 10% at 700 K until it reaches about 10.7% and it remains at that level for T_H greater than 1000 K.

By remarking that at higher heater temperature both H_2 and He gases generate almost same amount of power; besides at temperature greater than 1000 K the efficiency of H_2 gas remains around 10.7% while in He gas it exceeds 16%. Thus, at atmospheric charge pressure and high T_H, it is conclusively recommended to use He gas as the working fluid. Nevertheless, at low heater temperature ($T_H < 1000$ K) H_2 gas generates power greater by 10% than the power generated by He gas.

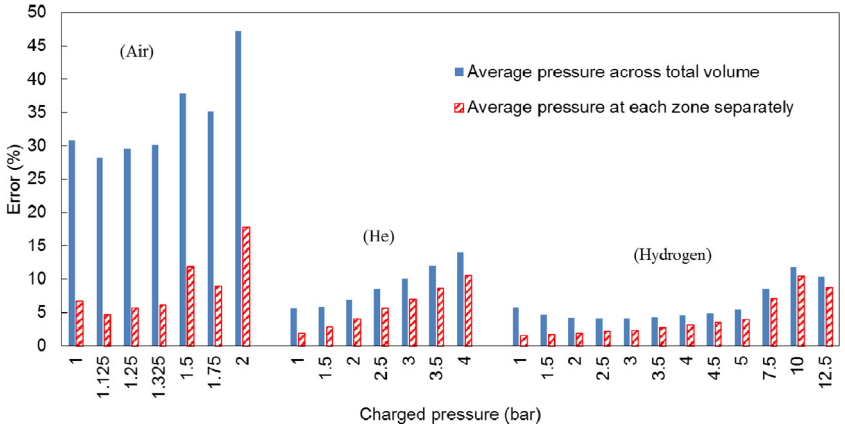


Fig. 9. Error of power output due to taking average pressure at each chamber separately.

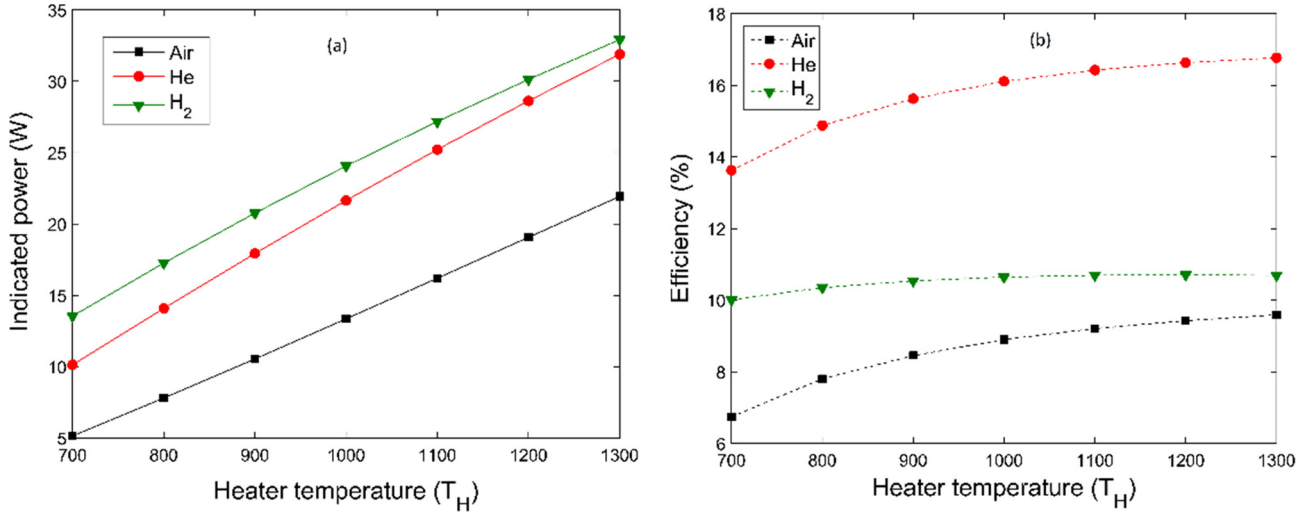


Fig. 10. Effects of T_H on Stirling engine performance at 1 bar charged pressure and T_C of 300 K; (a) work output and (b) efficiency.

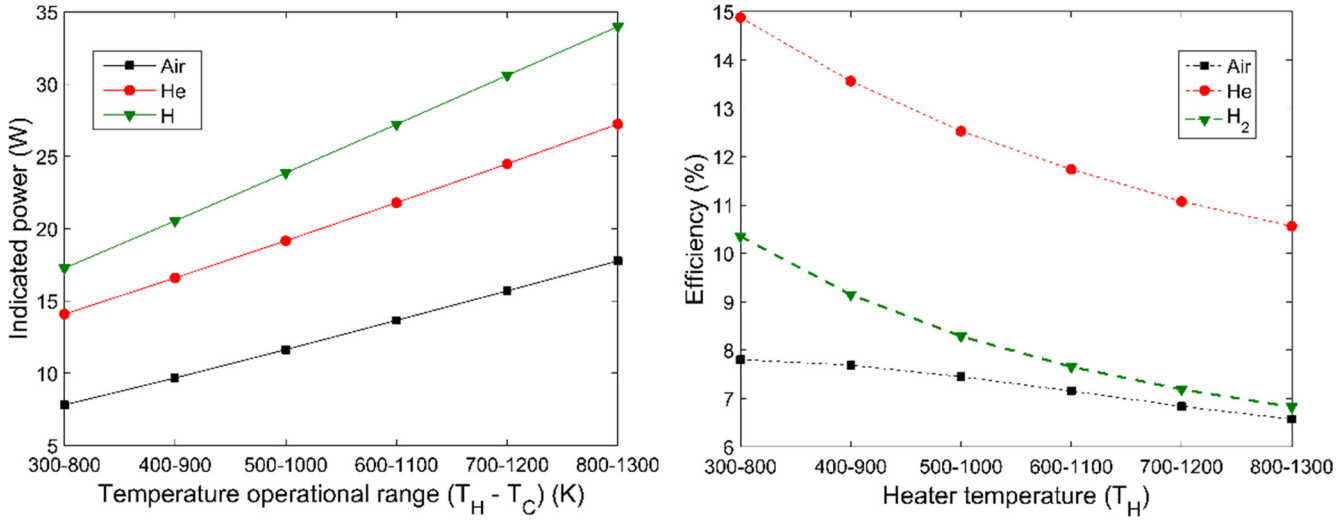


Fig. 11. Effect of shifting operational temperature range on Stirling engine performance at 1 bar charged pressure; (a) work output and (b) efficiency.

5.3. Effect of shifting temperature limits (T_H & T_C)

In this part, the temperature difference ($T_H - T_C$) are kept constant at a value of 500 K. Fig. 11a shows power output corresponding to operational temperature limits of (300–800 K) which have been shifted by step of 100 K until it reaches 800 K and 1300 K for T_C and T_H, respectively. The indicated power profiles are almost linear for all of the three types of working fluid. It is similar to the previous section as operational temperature range shifted to a higher level, thus a large amount of heat would be transferred to Stirling engine. As a result, more power output could be extracted. In terms of efficiency, shifting the operational temperature range would result in a decrease in the cycle thermal efficiency as shown in Fig. 11b.

Additionally, Table 4 represents a comparison of the indicated powers produced at different T_H with different T_C. All cases were conducted at 1 bar charge pressure. The first column represents cases of shifting T_H to high values while keeping T_C at 300 K, which leads to rising up in ΔT. The second column represents a case of shifting operational temperature ranges to high values while keeping temperature difference constant. It has been noticed that using air and He produce larger power output as the differences between

Table 4
Comparison of indicated power at different temperature limits.

T_H	T_{C1}/T_{C2}	Air		He		H_2	
800	300/300	7.8	7.8	14.1	14.1	17.3	17.3
900	300/400	10.5	9.7	17.9	16.6	20.8	20.5
1000	300/500	13.4	11.6	21.7	19.2	24.1	23.9
1100	300/600	16.2	13.7	25.2	21.8	27.2	27.2
1200	300/700	19.1	15.7	28.6	24.5	30.1	30.6
1300	300/800	21.9	17.8	31.9	27.2	32.9	34.0

temperature limits (T_H and T_C) increases. For example, at T_H equal to 1100 K; He gas produce 21.8 W (at T_C = 600 K) while when temperature difference increased the engine produced 25.2 W (at T_C = 300 K) for same source temperature. In contrast, hydrogen gas reflects different behavior. It shows slight improvement in the power output as ΔT increases up to 700 K. As ΔT becomes greater than 700 K the generated power becomes less than that obtained from ΔT = 500 K. For an operational range of (300–1300 K) and (800–1300 K), the first range produces 32.9 W while the later produces 34 W.

6. Conclusions

In this work, a comprehensive parametric study comprising 60 simulations of laminar flow for Stirling engine was performed. The study was conducted numerically on a β-type Stirling engine. The three operational factors were examined successfully and they are; initial charge pressure, thermal boundary condition and working fluid type. The following conclusions were withdrawn from this comprehensive study;

- There is small pressure variation across engine chambers and this leads to a disagreement between the net heat transfer and power output. This deviation comes as power output calculated from a p-v diagram by utilizing the average pressure through the whole volume. However, this deviation was reduced by dividing the whole volume into small sections. Then, the average pressure is calculated and used for each of the small sections separately instead of taking average pressure for the whole volume.
- The optimum charge pressure, which produces the most power output varies from a gas to another. For operational temperature-end-limits 300–800 K, these optimum pressures are; 1.75 bar (for air), 2.5 bar (for He) and 10 bar (for hydrogen).
- For H_2 and He gasses, particularly, it is preferred to run Stirling engine at a pressure less than optimum charge pressure of power output. Accordingly, thermal efficiency would be improved, while in return small amount of power output would be lost.
- At atmospheric charge pressure with a temperature limits difference (ΔT) greater than 700 K, it is recommended to use He gas to operate Stirling engine because it would provide both high efficiency and large power output. While at low temperature difference, hydrogen gas generate more power than He but still He gas has the best efficiency.
- Shifting the operational temperature range to high level increases the power output but it has a negative impact on thermal efficiency.
- For air and He gasses, the power output increases proportionally with temperature differences between T_H and T_C. While for hydrogen gas, the relation is proportionally up to ΔT equal to 700 K after that it shows reverse impacts.

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